



29th December 2024

**Economic Data Releases &
Their Historical Impact on
Assets**

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Quantwater Tech Investments Data Release Deck –29th December 2024

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Brief

Date	Day	Time (IST)	Cur.	Imp.	Event
30-Dec-24	Mon	20:15	USD	3	USD Chicago PMI (Dec)
31-Dec-24	Tue	5:30	Global	3	Major Markets closed or closing early
31-Dec-24	Tue	7:00	CNY	3	CNY Manufacturing PMI (Dec)
01-Jan-25	Wed	5:30	Global	3	Major Markets closed
02-Jan-25	Thur	5:30	Global	3	Markets closed: Swiss, Russia, NZ, Japan
02-Jan-25	Thur	19:00	USD	3	USD Initial Jobless Claims
02-Jan-25	Thur	20:15	USD	3	USD Manufacturing PMI (Dec)
02-Jan-25	Thur	20:30	USD	3	USD ISM Manufacturing PMI
02-Jan-25	Thur	20:30	USD	3	USD ISM Manufacturing Prices (Dec)
02-Jan-25	Thur	21:30	Global	3	Crude Oil Inventories
03-Jan-25	Thur	5:30	Global	3	Markets closed: Japan, Russia
04-Jan-25	Fri	20:30	USD	3	USD ISM Manufacturing PMI
04-Jan-25	Fri	20:30	USD	3	USD ISM Manufacturing Prices (Dec)
30-Jan-25	Thu	0:30	USD	3	Fed Interest Rate Decision
End					Annexure: Key Terms & Links

USD Chicago PMI (Dec)



USD Chicago PMI (Dec)

Event	USD Chicago PMI (Dec)			Description
Actual:				
Forecast:	42.7			
Previous:	40.2			
Time (GMT):	14:45			
Release Date	Actual	Forecast	Previous	The Chicago Purchasing Managers' Index (PMI) is a diffusion index incorporating survey results provided by manufacturing firms around the Chicago area. A reading above fifty suggests the manufacturing sector is expanding, while a reading below fifty suggests the manufacturing sector is in contraction. The Chicago PMI can be of some assistance in forecasting the ISM manufacturing PMI. A reading that is stronger than forecast is generally supportive (bullish) for the USD, while a weaker than forecast reading is generally negative (bearish) for the USD. -Historical Impact on Assets: EURUSD USDJPY
29-Nov-24	40.2	44.9	41.6	
31-Oct-24	41.6	46.9	46.6	
30-Sep-24	46.6	46.1	46.1	
30-Aug-24	46.1	45.0	45.3	
31-Jul-24	45.3	44.8	47.4	
28-Jun-24	47.4	39.7	35.4	

USD Chicago PMI (Dec)
EURUSD



C:EURUSD Price Action - 2024-06-28 13:45 GMT



- ✖ Event Time
- - - +1 Volatility Unit
- - - -1 Volatility Unit
- ✕ Max Peak
- ✕ Min Peak

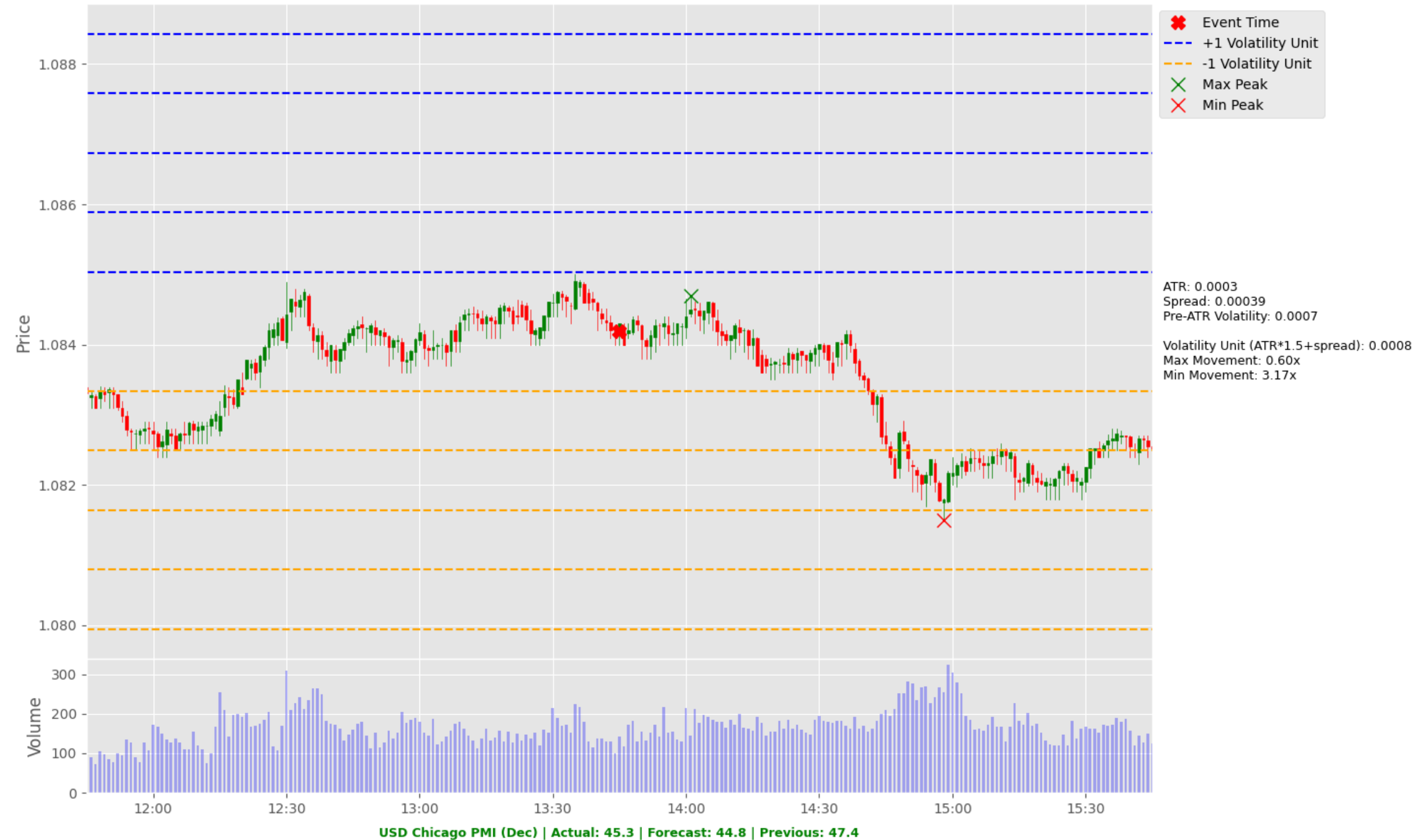
ATR: 0.0004
Spread: 0.00039
Pre-ATR Volatility: 0.0012

Volatility Unit (ATR*1.5+spread): 0.0010
Max Movement: 2.12x
Min Movement: 1.00x



USD Chicago PMI (Dec) | Actual: 47.4 | Forecast: 39.7 | Previous: 35.4

C:EURUSD Price Action - 2024-07-31 13:45 GMT



C:EURUSD Price Action - 2024-08-30 13:45 GMT



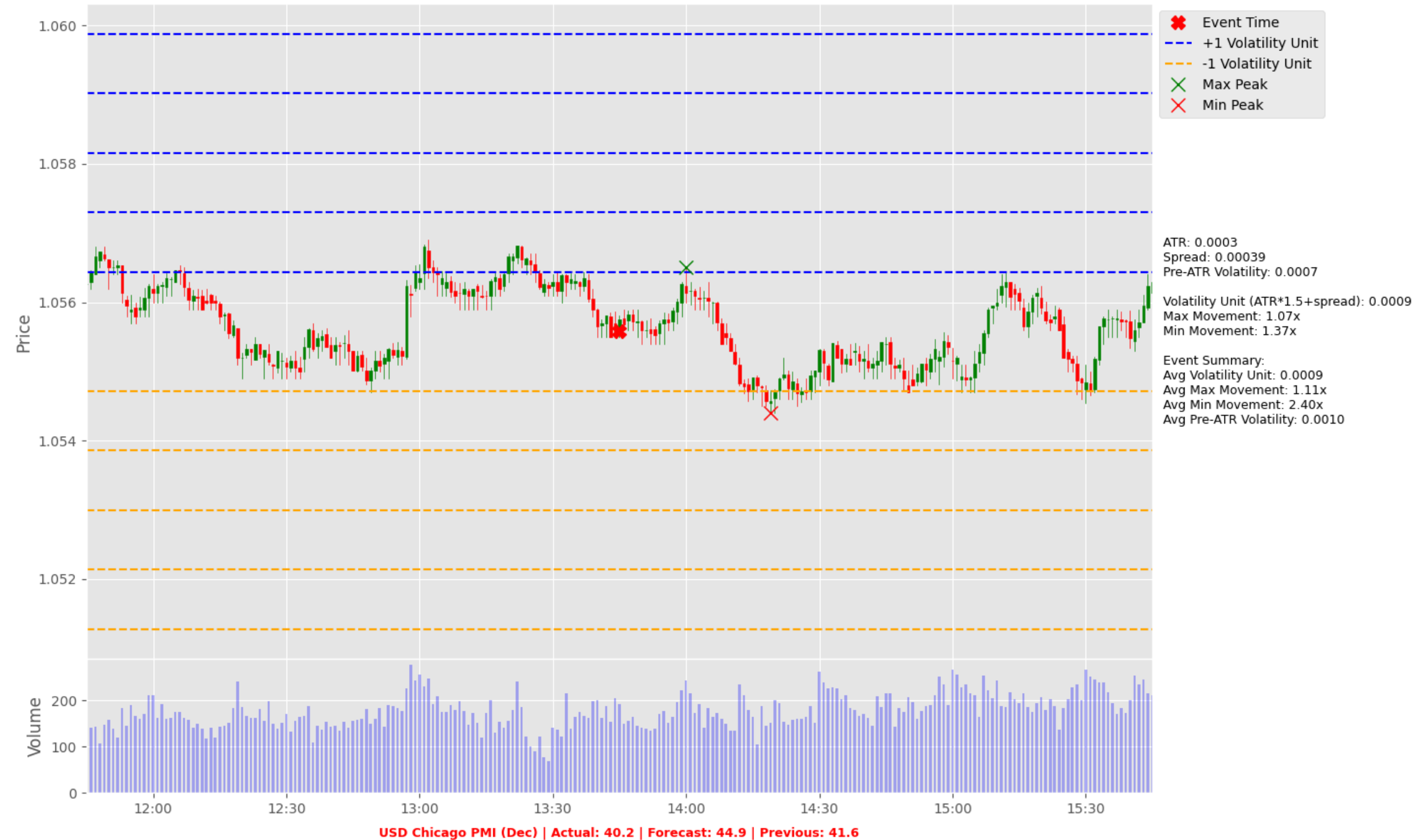
C:EURUSD Price Action - 2024-09-30 13:45 GMT



C:EURUSD Price Action - 2024-10-31 13:45 GMT



C:EURUSD Price Action - 2024-11-29 13:45 GMT



USD Chicago PMI (Dec)
USDJPY



C:USDJPY Price Action - 2024-06-28 13:45 GMT



- ✖ Event Time
- - - +1 Volatility Unit
- - - -1 Volatility Unit
- ✕ Max Peak
- ✕ Min Peak

ATR: 0.0485
Spread: 0.09900
Pre-ATR Volatility: 0.1600

Volatility Unit (ATR*1.5+spread): 0.1717
Max Movement: 3.76x
Min Movement: 0.27x

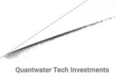
Price

Volume



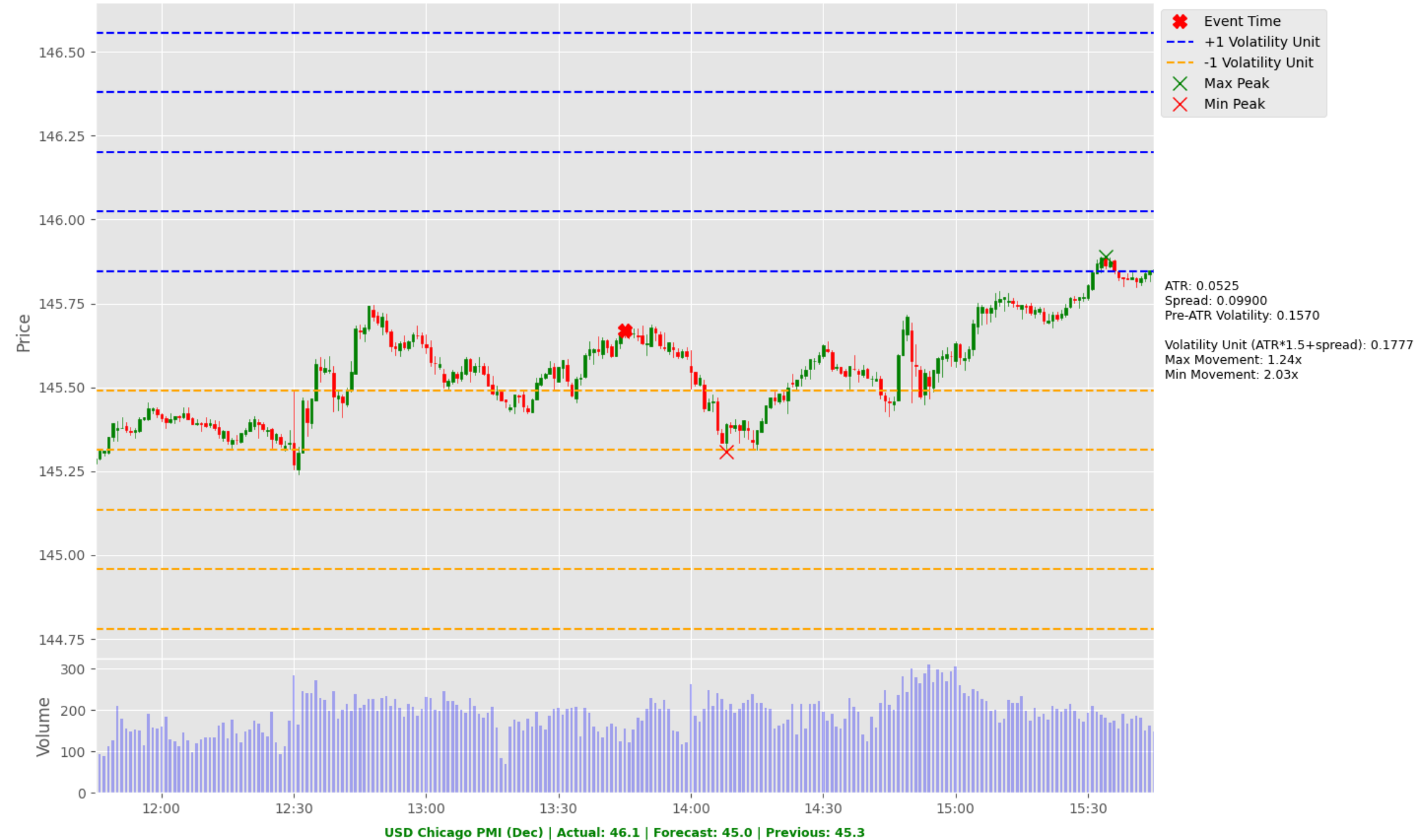
USD Chicago PMI (Dec) | Actual: 47.4 | Forecast: 39.7 | Previous: 35.4

C:USDJPY Price Action - 2024-07-31 13:45 GMT

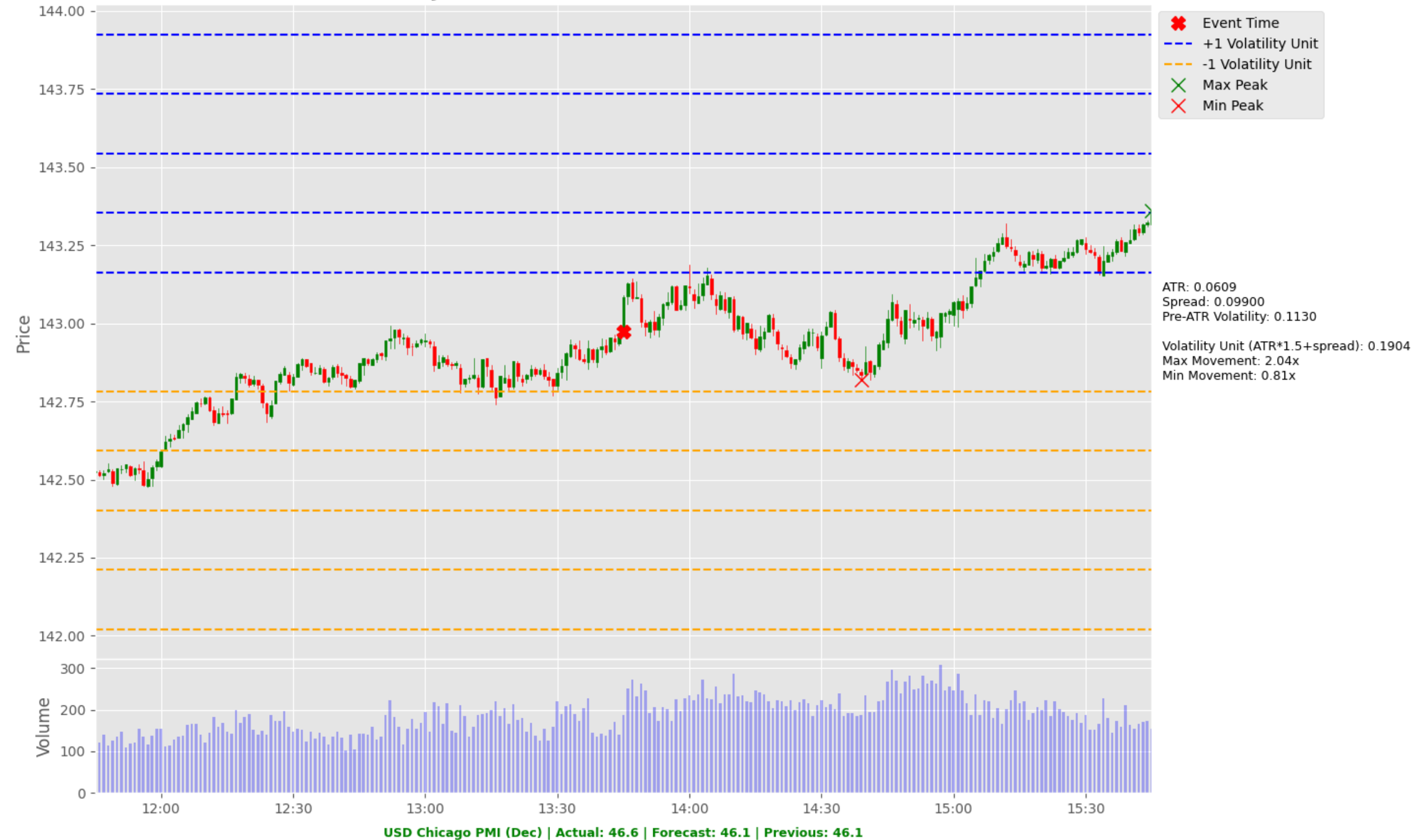


USD Chicago PMI (Dec) | Actual: 45.3 | Forecast: 44.8 | Previous: 47.4

C:USDJPY Price Action - 2024-08-30 13:45 GMT



C:USDJPY Price Action - 2024-09-30 13:45 GMT



C:USDJPY Price Action - 2024-10-31 13:45 GMT



C:USDJPY Price Action - 2024-11-29 13:45 GMT



CNY Manufacturing PMI (Dec)



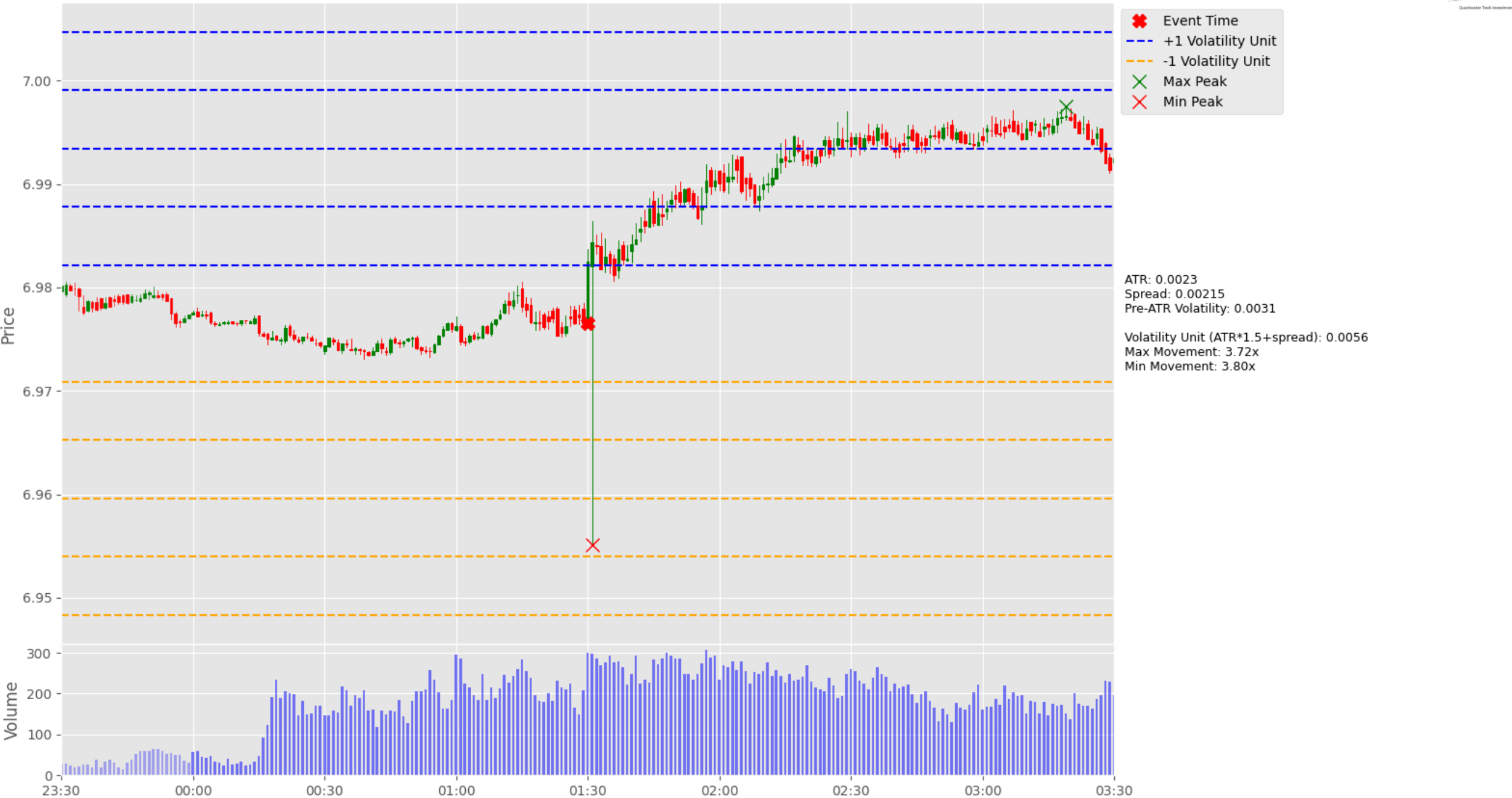
CNY Manufacturing PMI (Dec)

Event	CNY Manufacturing PMI (Dec)			Description
Actual:				China Manufacturing Purchasing Managers Index (PMI) provides an early indication each month of economic activities in the Chinese manufacturing sector. It is compiled by China Federation of Logistics & Purchasing (CFLP) and China Logistics Information Centre (CLIC), based on data collected by the National Bureau of Statistics (NBS).Li & Fung Research Centre is responsible for drafting and disseminating the English PMI report. Every month questionnaires are sent to over 700 manufacturing enterprises all over China. The data presented here is compiled from the enterprises responses about their purchasing activities and supply situations. The PMI should be compared to other economic data sources when used in decision-making. A reading that is stronger than forecast is generally supportive (bullish) for the CNY , while a weaker than forecast reading is generally negative (bearish) for the CNY. -Historical Impact on Assets: USDCNH USDCNY
Forecast:	50.3			
Previous:	50.3			
Time (GMT):	1:30			
Release Date	Actual	Forecast	Previous	
30-Nov-24	50.3	50.2	50.1	
31-Oct-24	50.1	49.8	49.8	
30-Sep-24	49.8	49.4	49.1	
31-Aug-24	49.1	49.5	49.4	
31-Jul-24	49.4	49.4	49.5	
30-Jun-24	49.5	49.5	49.5	

CNY Manufacturing PMI (Dec)
USDCNH

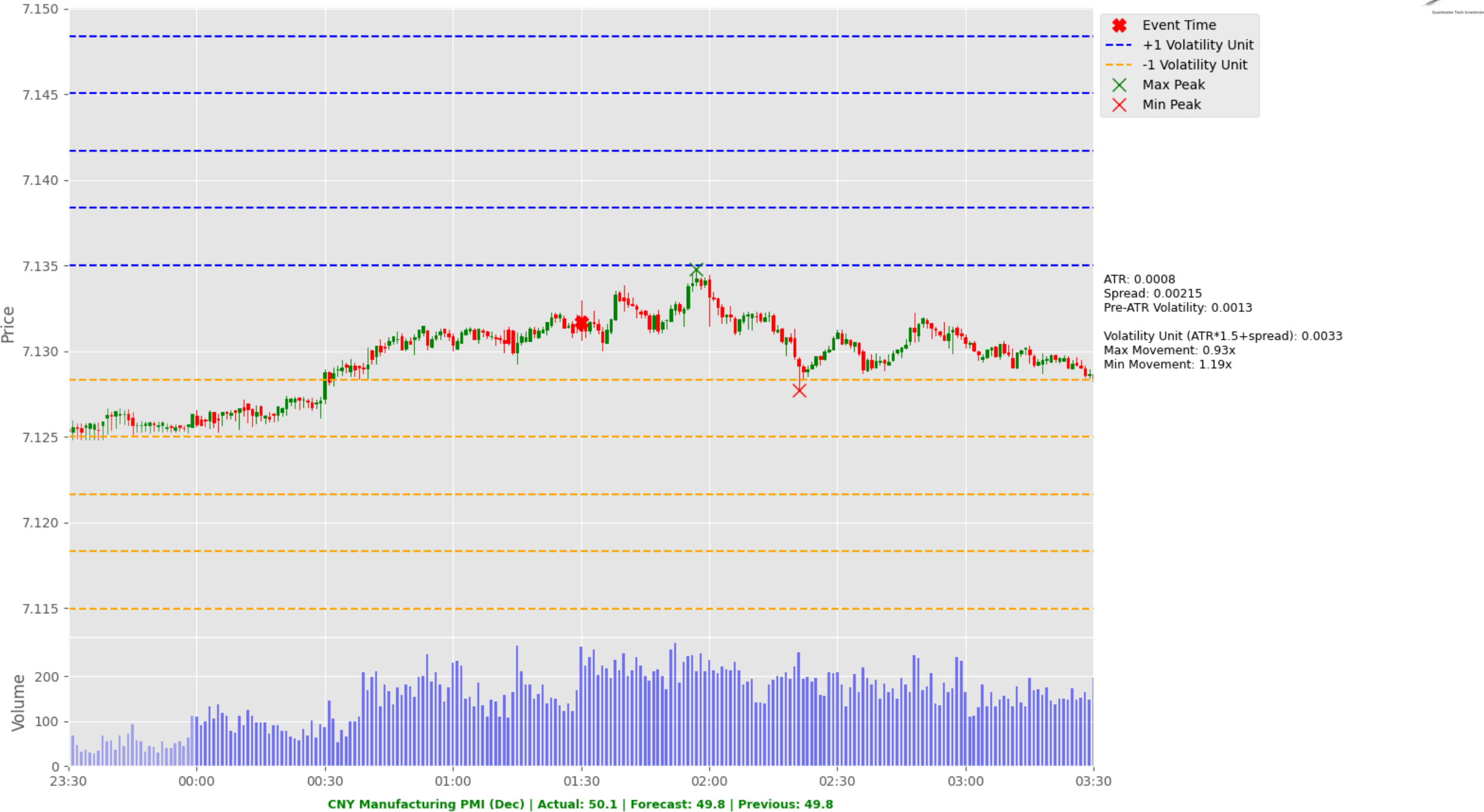
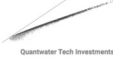


C:USDCNH Price Action - 2024-09-30 01:30 GMT



CNY Manufacturing PMI (Dec) | Actual: 49.8 | Forecast: 49.4 | Previous: 49.1

C:USDCNH Price Action - 2024-10-31 01:30 GMT



USD Initial Jobless Claims



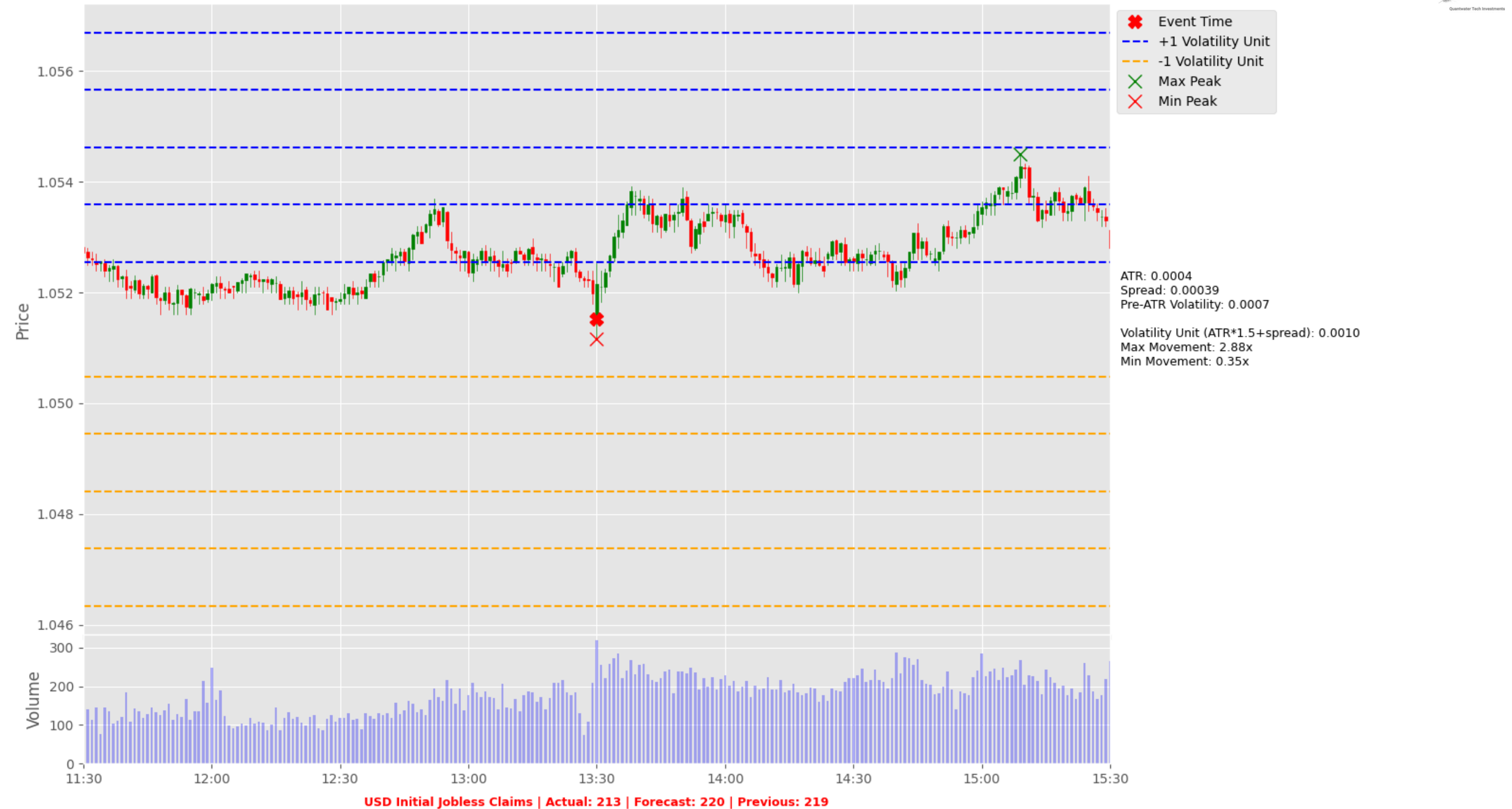
USD Initial Jobless Claims

Event	USD Initial Jobless Claims			Description
Actual:				Initial Jobless Claims measures the number of people who filed for unemployment insurance for the first time during the past week. This is the most timely U.S. economic data, but the market impact varies from week to week. A reading that is higher than forecast is generally negative (bearish) for the USD, while a lower than forecast reading is generally supportive (bullish) for the USD. -Historical Impact on Assets: EURUSD USDJPY
Forecast:	220K			
Previous:	219K			
Time (GMT):	13:30			
Release Date	Actual	Forecast	Previous	
26-Dec-24	219	223	220	
19-Dec-24	220	229	242	
12-Dec-24	242	221	225	
05-Dec-24	224	215	215	
27-Nov-24	213	215	215	
21-Nov-24	213	220	219	

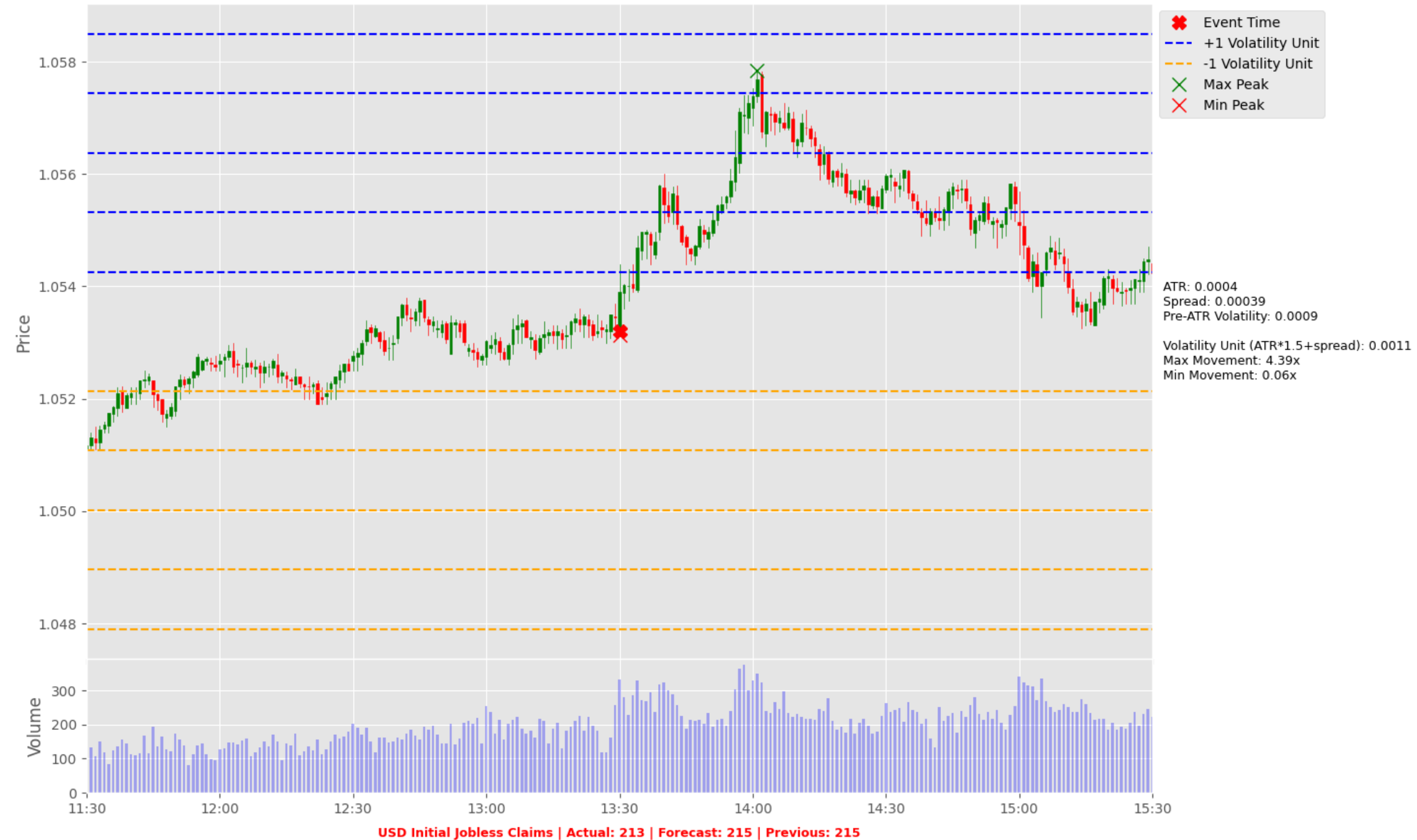
USD Initial Jobless Claims
EURUSD



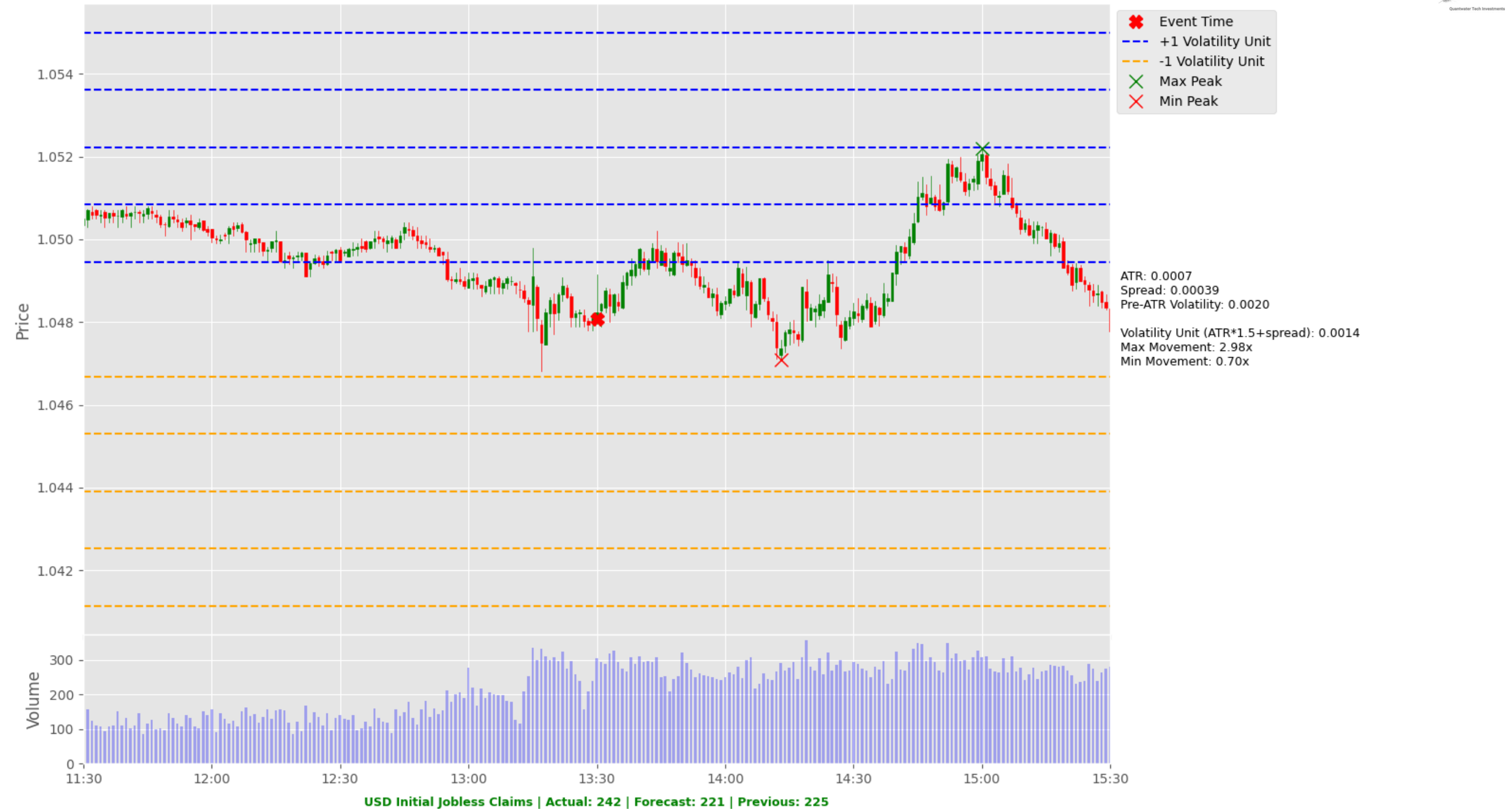
C:EURUSD Price Action - 2024-11-21 13:30 GMT



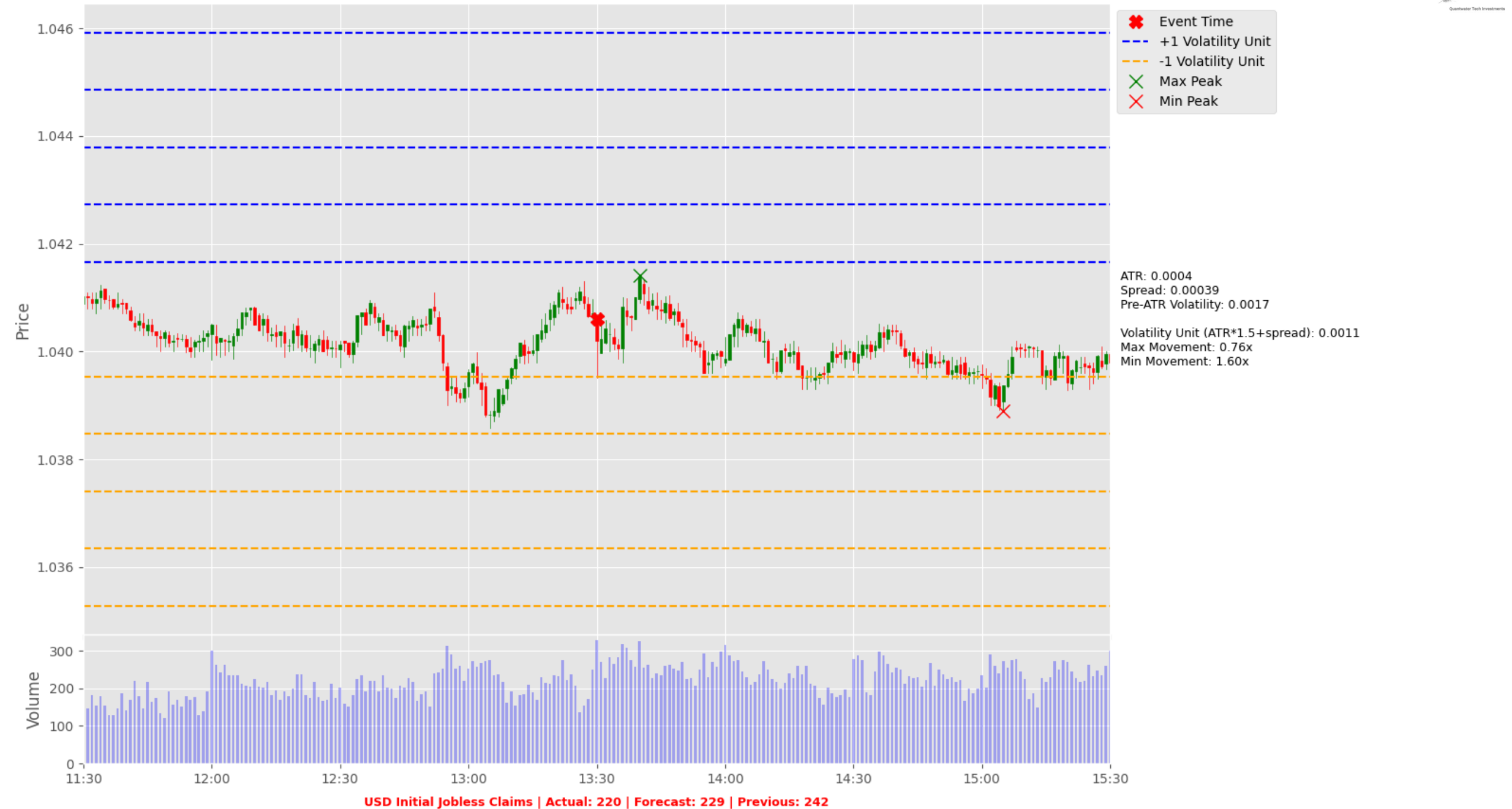
C:EURUSD Price Action - 2024-11-27 13:30 GMT



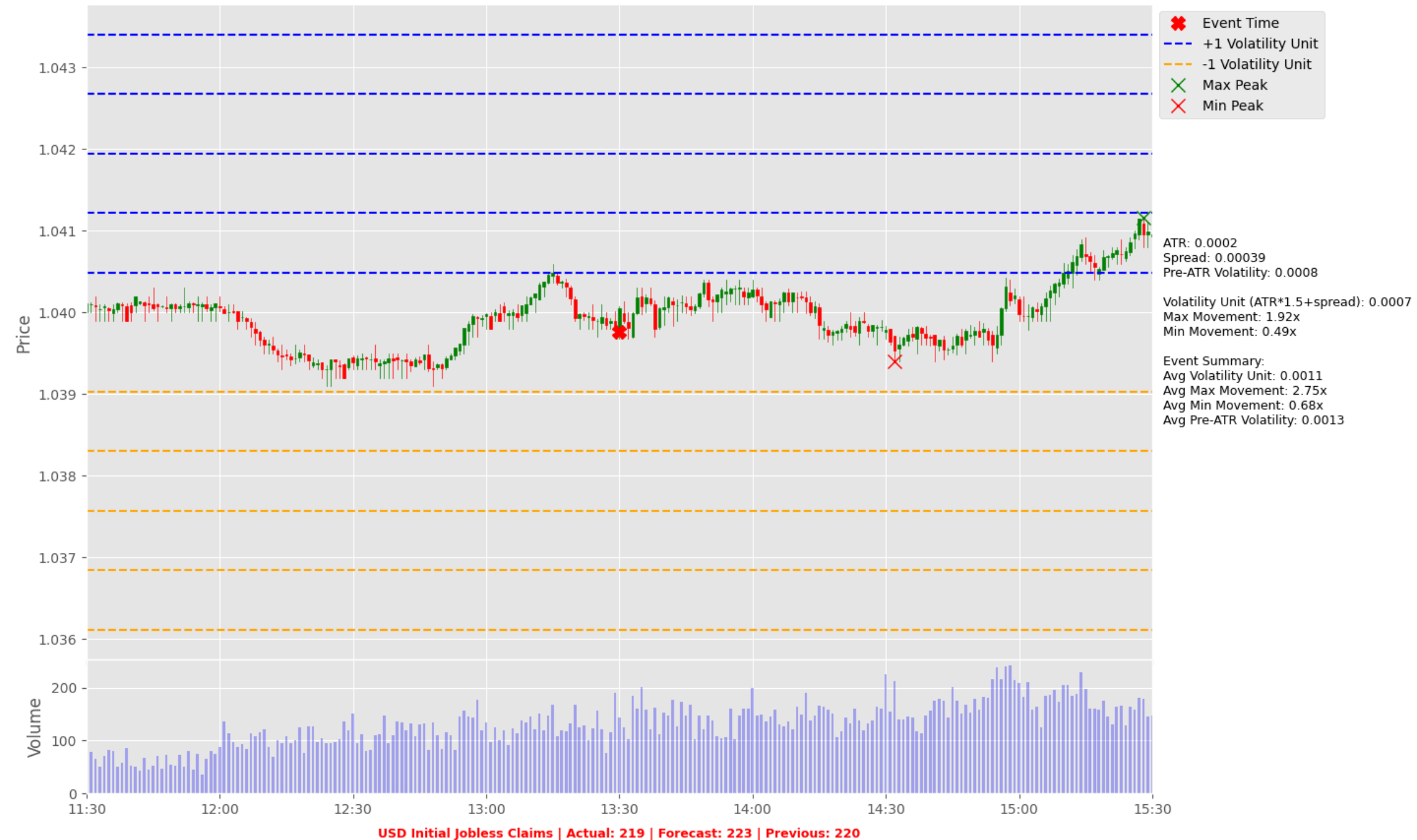
C:EURUSD Price Action - 2024-12-12 13:30 GMT



C:EURUSD Price Action - 2024-12-19 13:30 GMT



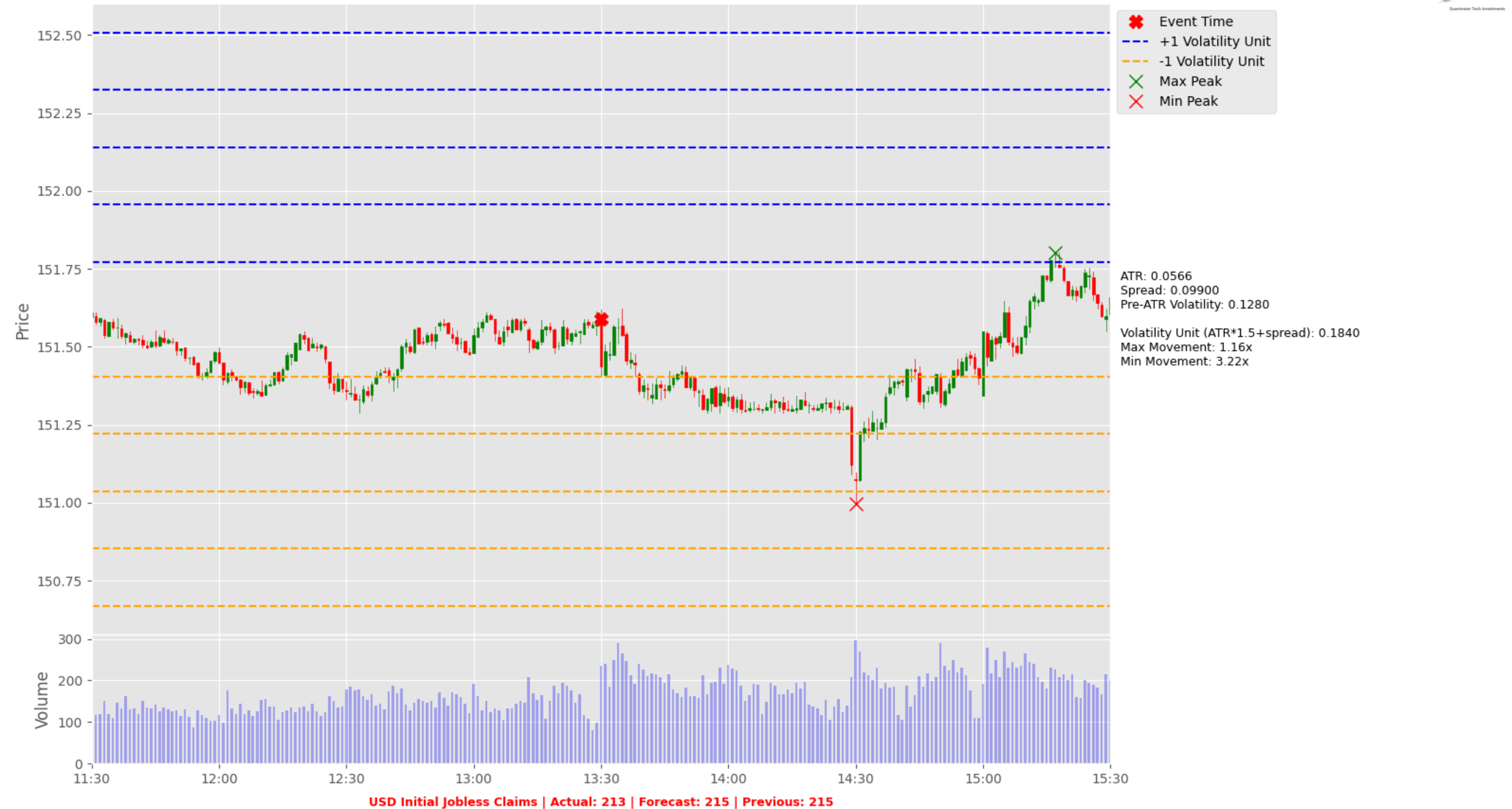
C:EURUSD Price Action - 2024-12-26 13:30 GMT



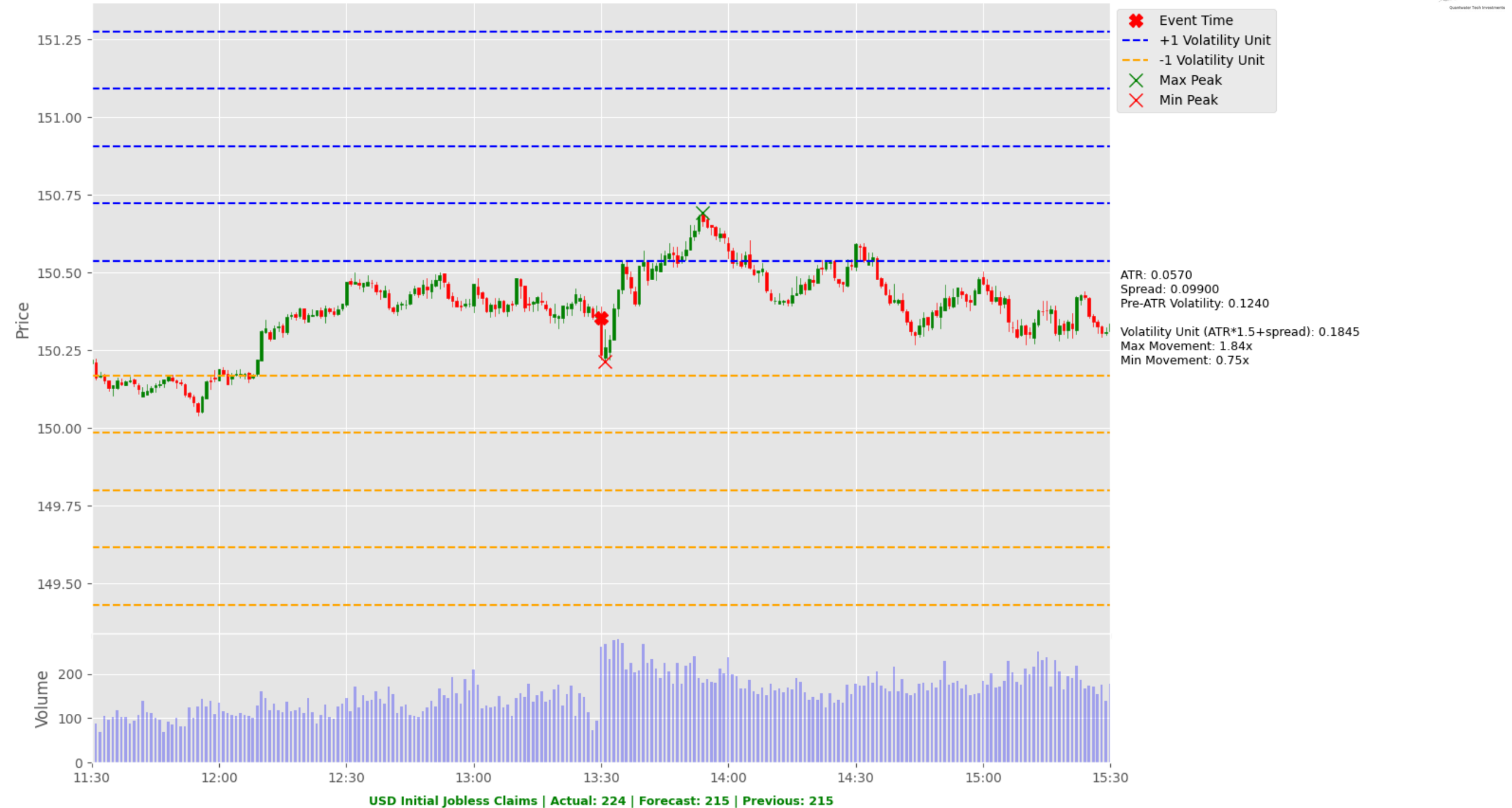
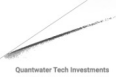
USD Initial Jobless Claims
USDJPY



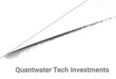
C:USDJPY Price Action - 2024-11-27 13:30 GMT



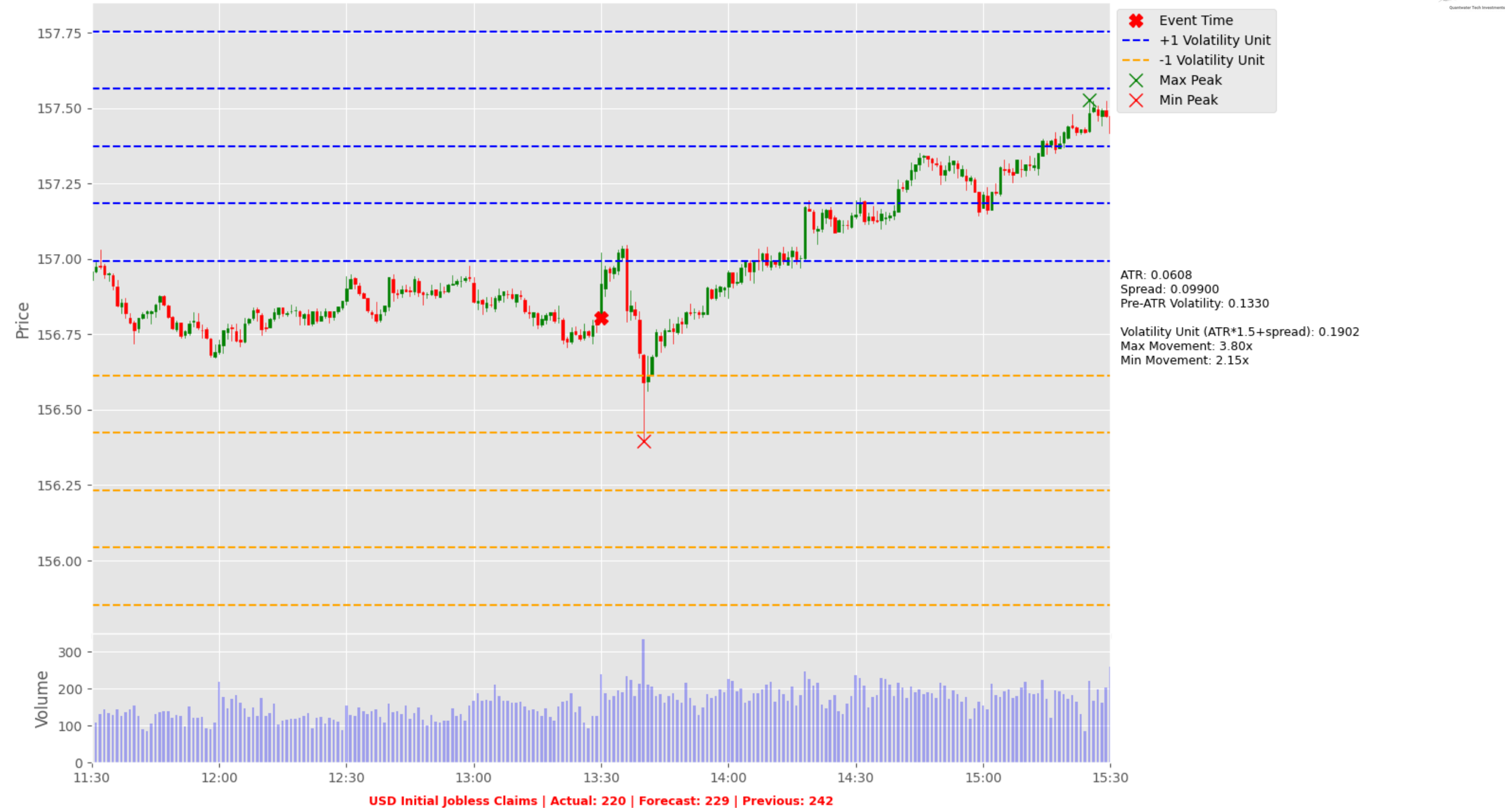
C:USDJPY Price Action - 2024-12-05 13:30 GMT



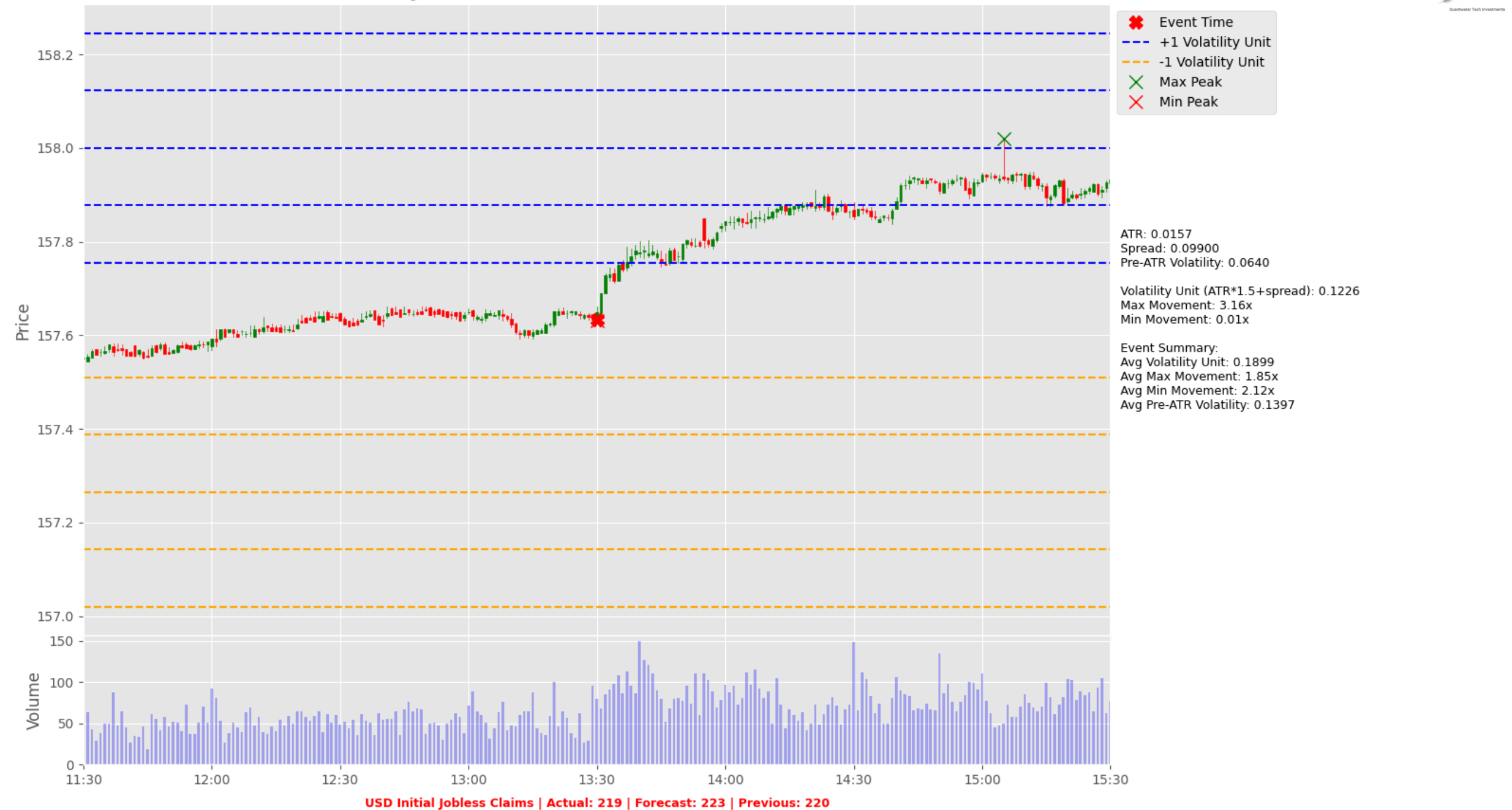
C:USDJPY Price Action - 2024-12-12 13:30 GMT



C:USDJPY Price Action - 2024-12-19 13:30 GMT



C:USDJPY Price Action - 2024-12-26 13:30 GMT



USD Manufacturing PMI (Dec)



USD Manufacturing PMI (Dec)

Event	USD Manufacturing PMI (Dec)			Description
Actual:				The Manufacturing Purchasing Managers' Index (PMI) is a diffusion index incorporating survey results provided by manufacturing firms throughout the country. A reading above fifty suggests the manufacturing sector is expanding, while a reading below fifty suggests the manufacturing sector is in contraction. Policymakers and traders watch these surveys closely as purchasing managers usually have early access to data about their company's performance, rather than waiting for the hard data to emerge. While manufacturing accounts for a far smaller percentage of economic output than services, it is far more cyclical and is therefore a useful indicator of where in the economic cycle the economy is currently situated. A reading that is stronger than forecast is generally supportive (bullish) for the USD while a weaker than forecast reading is generally negative (bearish) for the USD. -Historical Impact on Assets: EURUSD USDJPY
Forecast:	48.30			
Previous:	49.70			
Time (GMT):	14:45			
Release Date	Actual	Forecast	Previous	
16-Dec-24	48.3	49.4	49.7	
02-Dec-24	49.7	48.8	48.5	
22-Nov-24	48.8	48.8	48.5	
01-Nov-24	48.5	47.8	47.8	
24-Oct-24	47.8	47.5	47.3	
01-Oct-24	47.3	47.0	47.9	

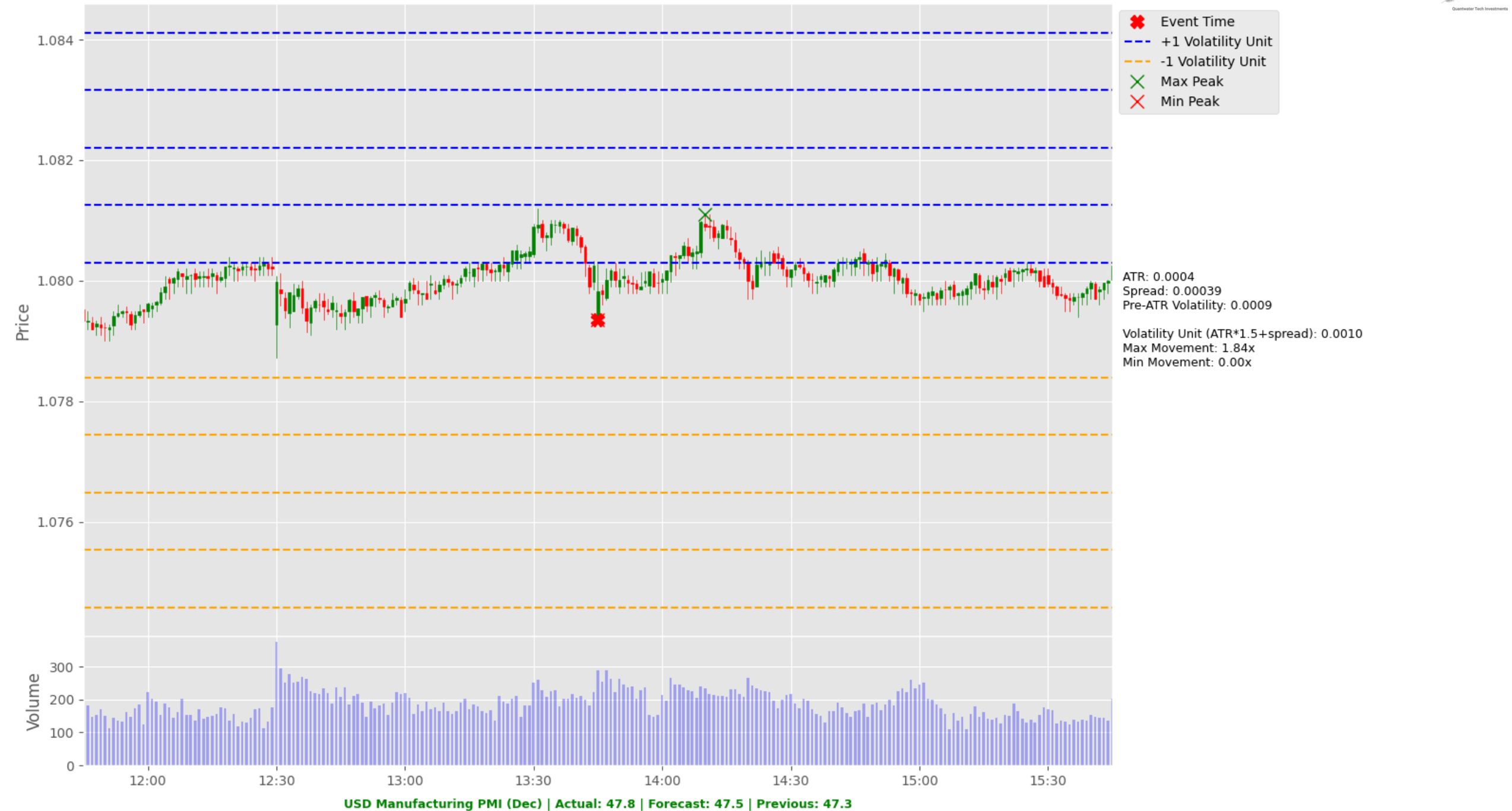
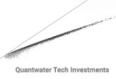
USD Manufacturing PMI (Dec)
EURUSD



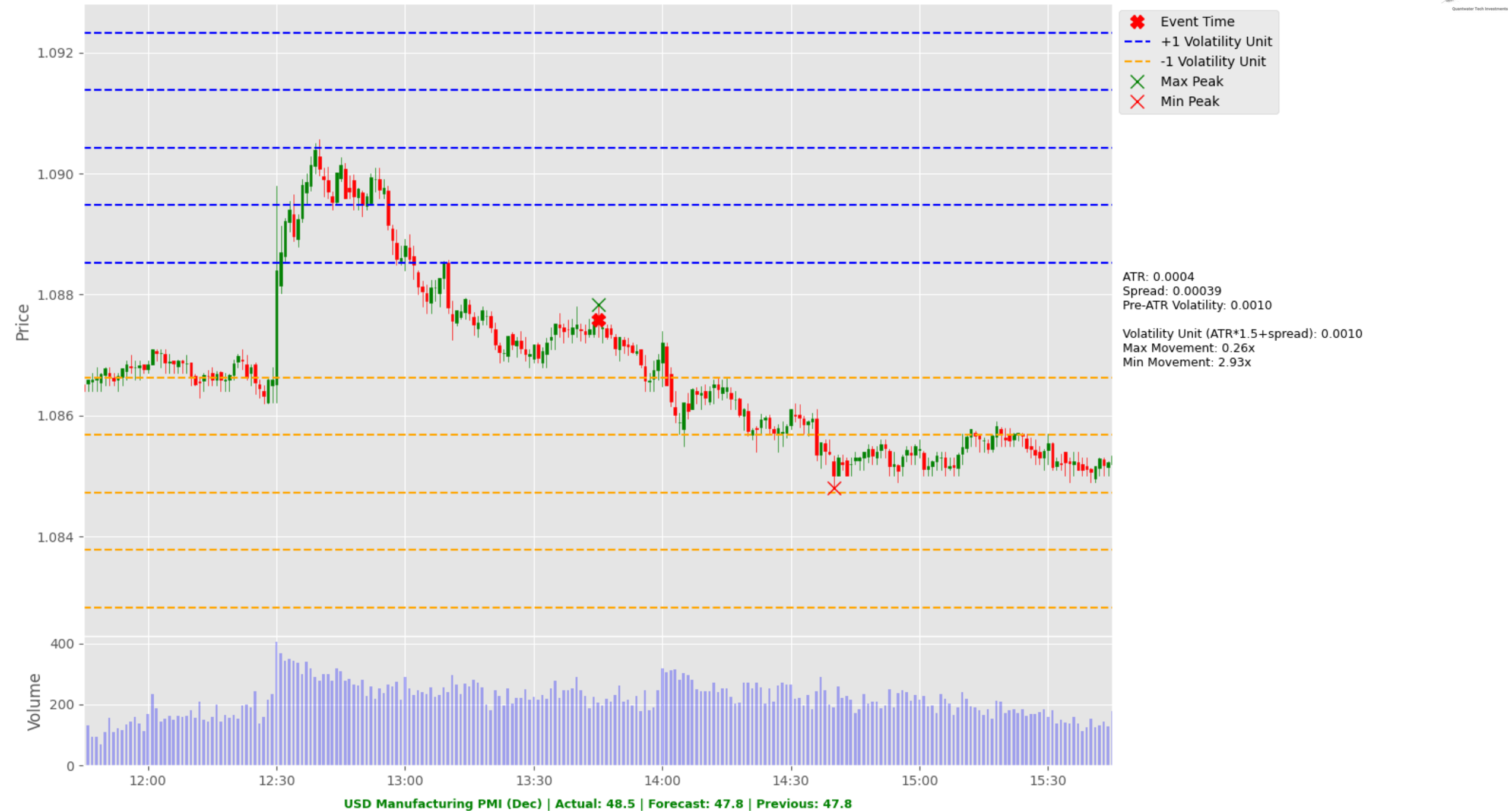
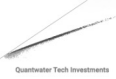
C:EURUSD Price Action - 2024-10-01 13:45 GMT



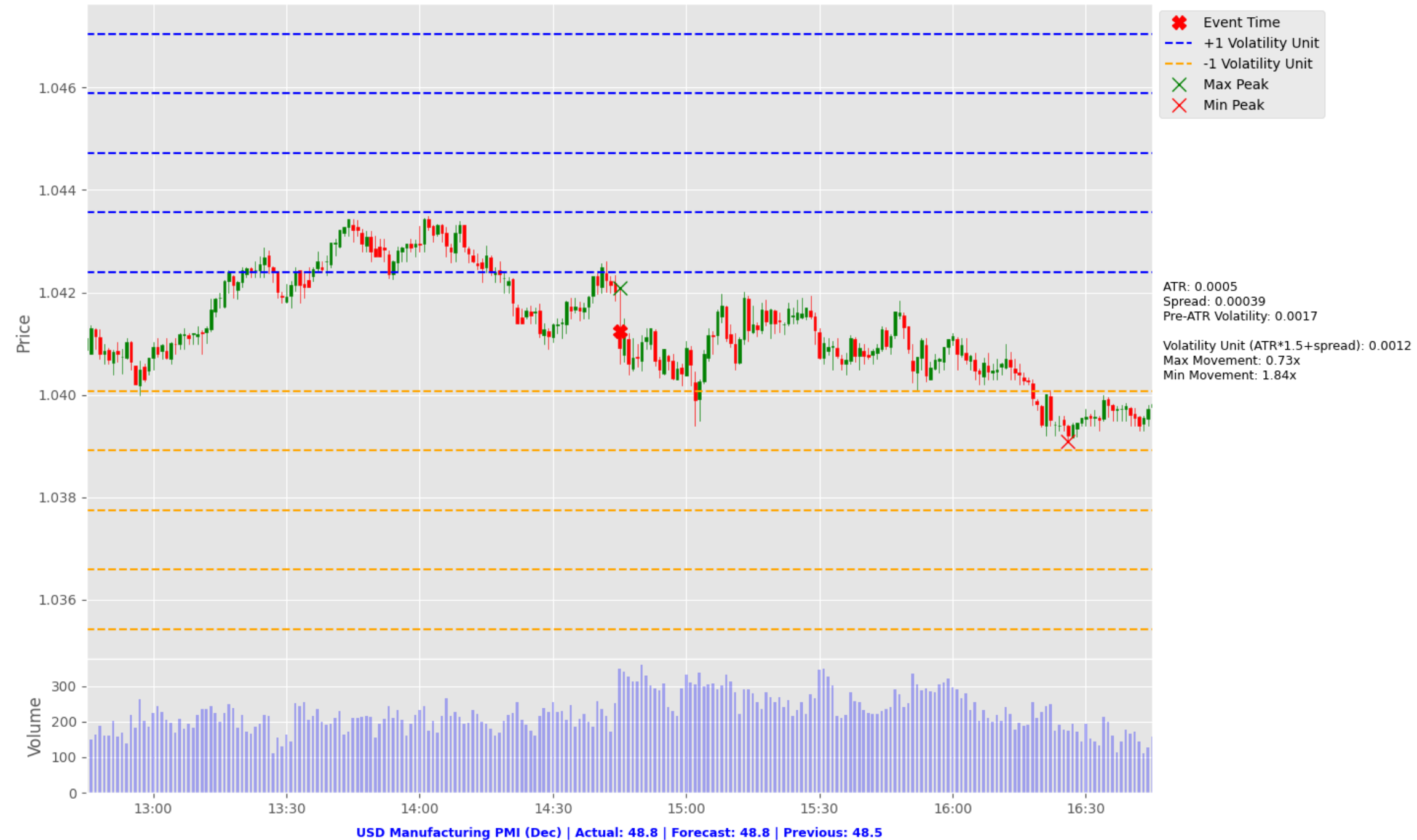
C:EURUSD Price Action - 2024-10-24 13:45 GMT



C:EURUSD Price Action - 2024-11-01 13:45 GMT



C:EURUSD Price Action - 2024-11-22 14:45 GMT



C:EURUSD Price Action - 2024-12-02 14:45 GMT



C:EURUSD Price Action - 2024-12-16 14:45 GMT



USD Manufacturing PMI (Dec)
USDJPY



C:USDJPY Price Action - 2024-07-01 14:00 GMT

Price

- ✖ Event Time
- - - +1 Volatility Unit
- - - -1 Volatility Unit
- ✕ Max Peak
- ✕ Min Peak

ATR: 0.0564
Spread: 0.09900
Pre-ATR Volatility: 0.1200

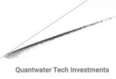
Volatility Unit (ATR*1.5+spread): 0.1835
Max Movement: 2.26x
Min Movement: 2.06x

Volume

USD ISM Manufacturing PMI | Actual: 48.5 | Forecast: 49.2 | Previous: 48.7



C:USDJPY Price Action - 2024-08-01 14:00 GMT



USD ISM Manufacturing PMI | Actual: 46.8 | Forecast: 48.8 | Previous: 48.5

C:USDJPY Price Action - 2024-09-03 14:00 GMT

Price

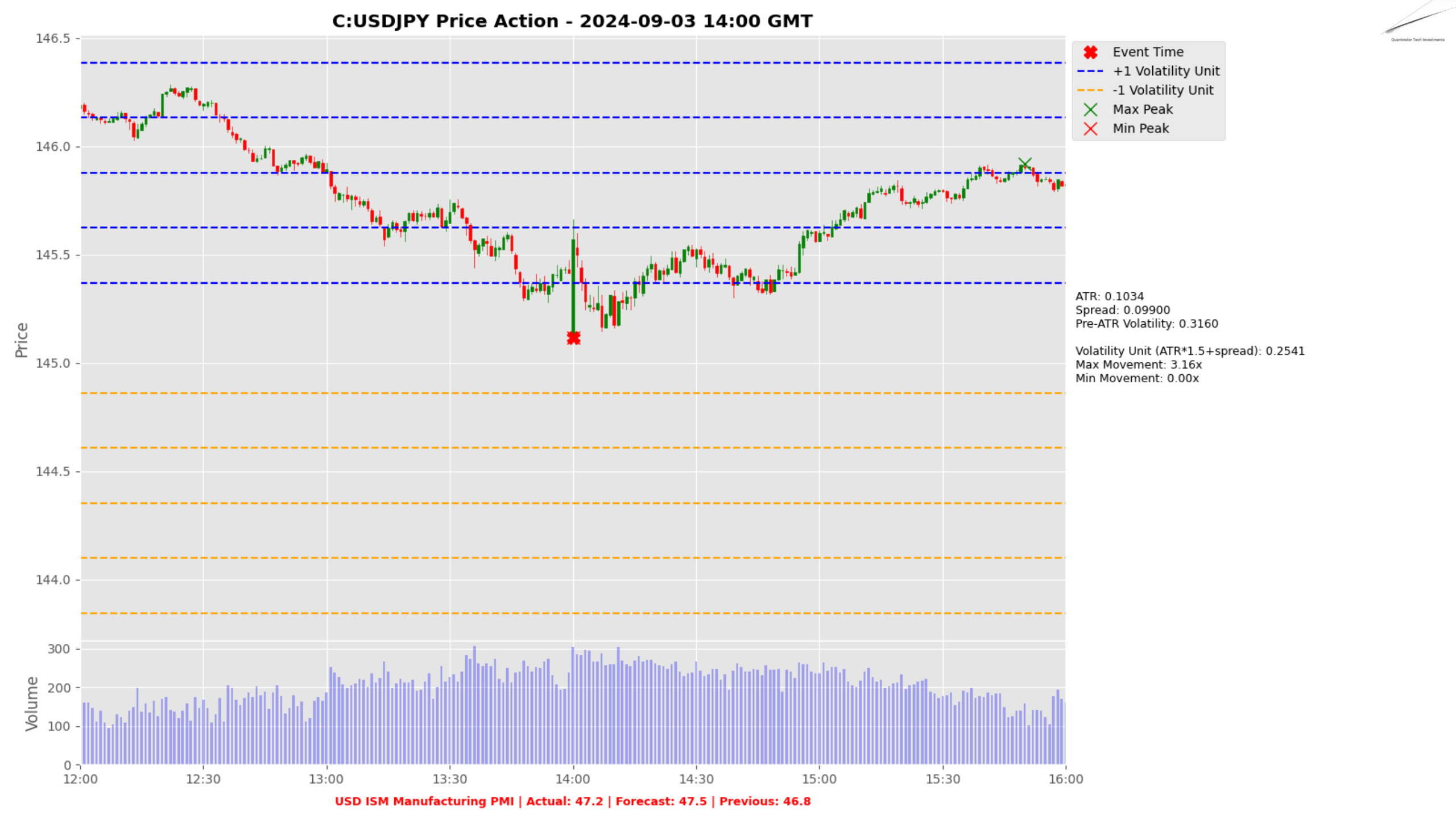
- ✖ Event Time
- - +1 Volatility Unit
- - -1 Volatility Unit
- ✕ Max Peak
- ✕ Min Peak

ATR: 0.1034
Spread: 0.09900
Pre-ATR Volatility: 0.3160

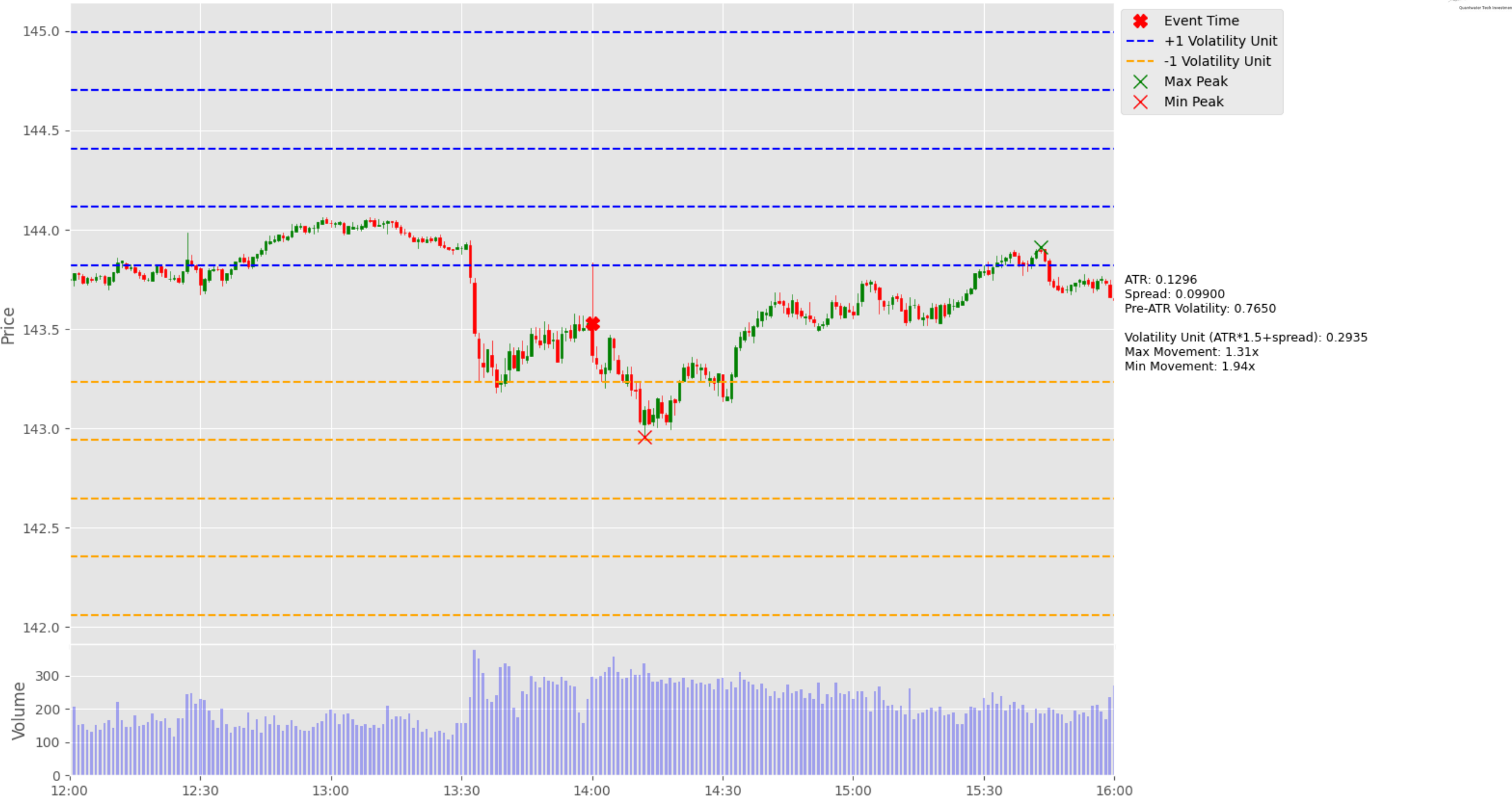
Volatility Unit (ATR*1.5+spread): 0.2541
Max Movement: 3.16x
Min Movement: 0.00x

Volume

USD ISM Manufacturing PMI | Actual: 47.2 | Forecast: 47.5 | Previous: 46.8

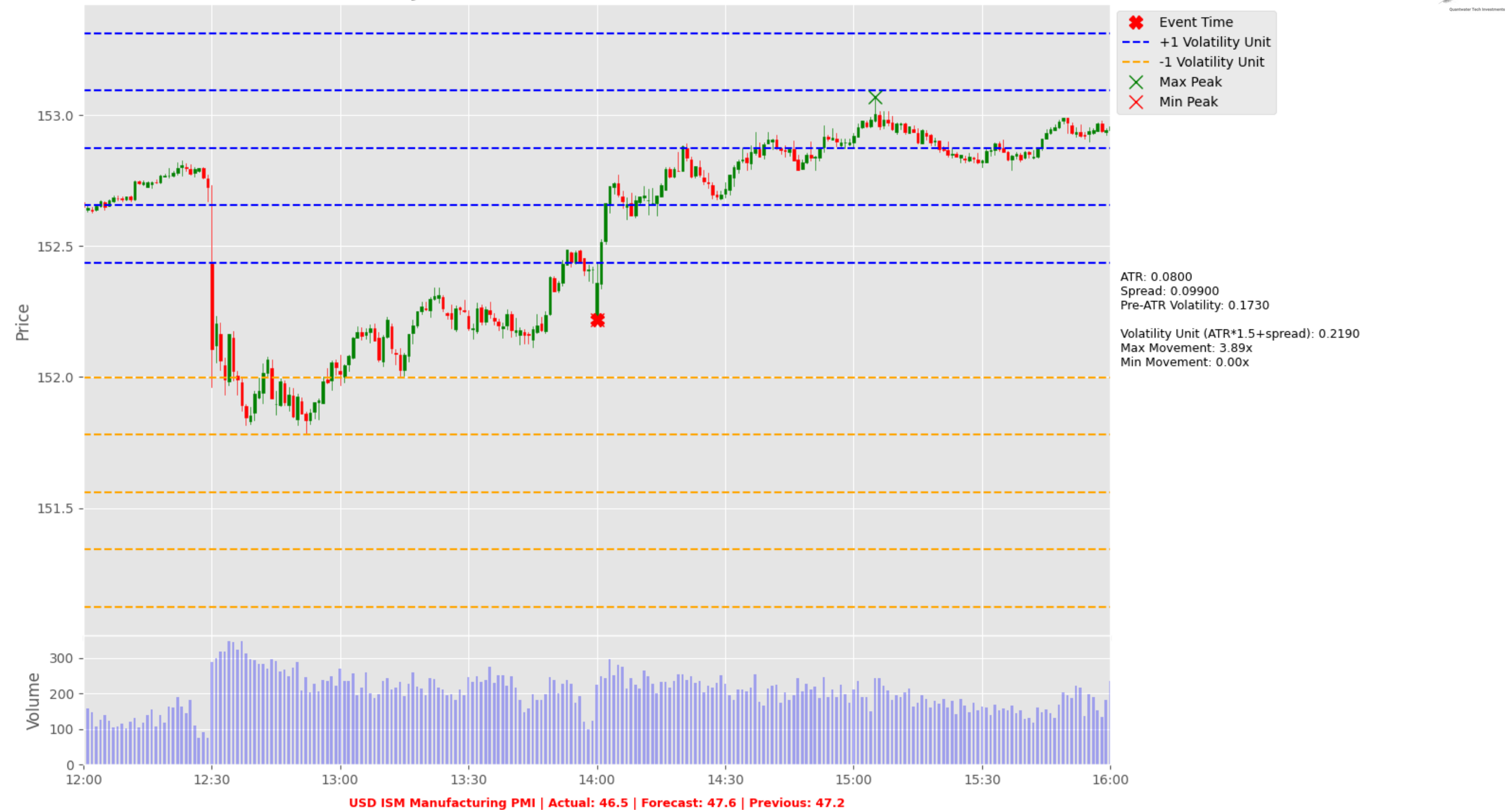


C:USDJPY Price Action - 2024-10-01 14:00 GMT

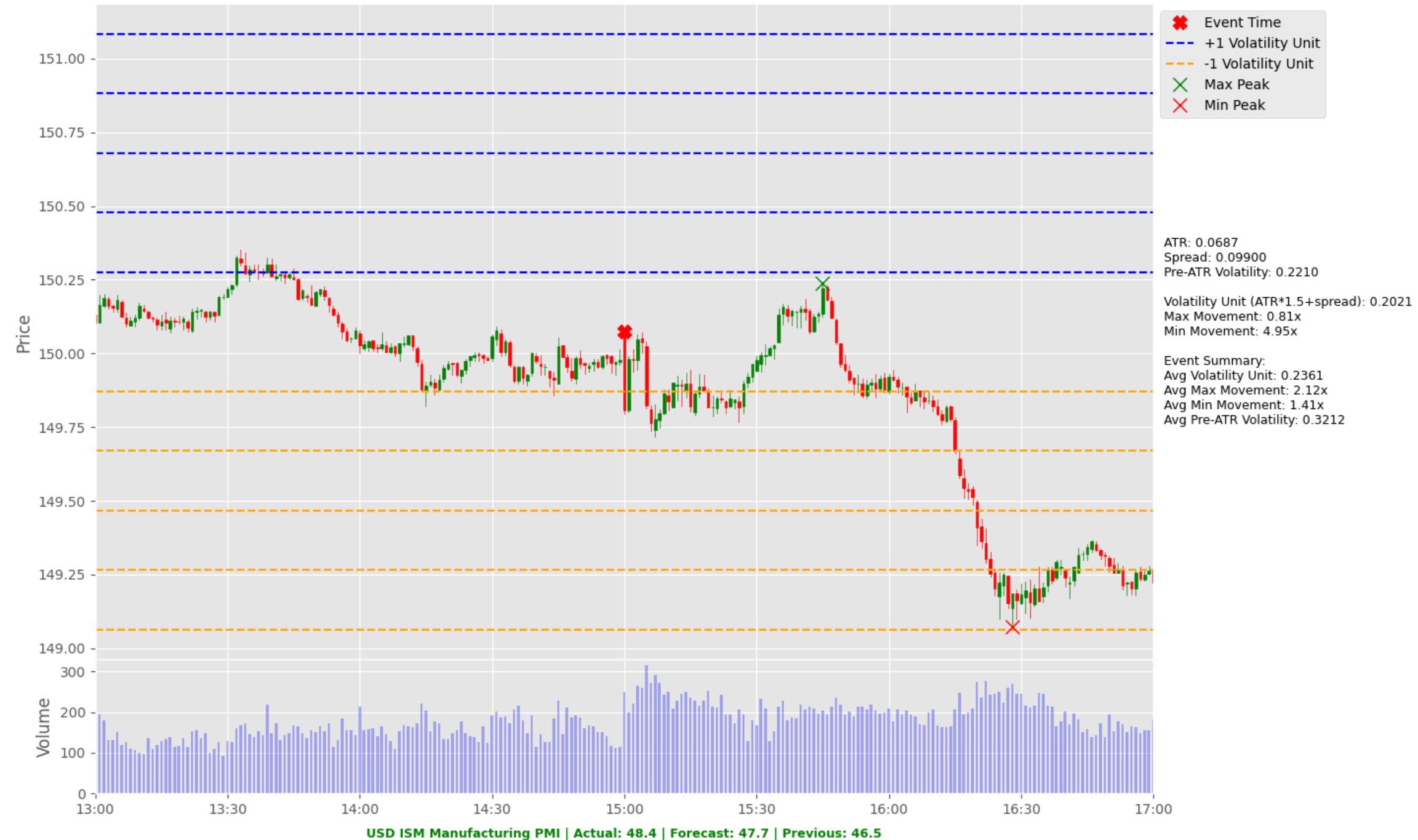


USD ISM Manufacturing PMI | Actual: 47.2 | Forecast: 47.6 | Previous: 47.2

C:USDJPY Price Action - 2024-11-01 14:00 GMT



C:USDJPY Price Action - 2024-12-02 15:00 GMT



USD ISM Manufacturing Prices (Dec)



USD ISM Manufacturing Prices (Dec)

Event	USD ISM Manufacturing Prices (Dec)			Description
Actual:				The Institute of Supply Management (ISM) Manufacturing Purchasing Managers Index (PMI) Report on Business is based on data compiled from monthly replies to questions asked of purchasing and supply executives in over 400 industrial companies. For each of the indicators measured (New Orders, Backlog of Orders, New Export Orders, Imports, Production, Supplier Deliveries, Inventories, Customers Inventories, Employment, and Prices), this report shows the percentage reporting each response, the net difference between the number of responses in the positive economic direction and the negative economic direction and the diffusion index. Responses are raw data and are never changed. The diffusion index includes the percent of positive responses plus one-half of those responding the same (considered positive). The resulting single index number is then seasonally adjusted to allow for the effects of repetitive intra-year variations resulting primarily from normal differences in weather conditions, various institutional arrangements, and differences attributable to non-moveable holidays. All seasonal adjustment factors are supplied by the U.S. Department of Commerce and are subject annually to relatively minor changes when conditions warrant them. The PMI is a composite index based on the seasonally adjusted diffusion indices for five of the indicators with varying weights: New Orders --30% Production --25% Employment --20% Supplier Deliveries --15% and Inventories -- 10%. The Prices Paid subcategory is a diffusion index calculated by adding the percent of responses indicating they paid more for inputs plus one-half of those responding that they paid the same for inputs. The resulting single index number is then seasonally adjusted. The Prices Paid diffusion index is one of a number of indicators pointing to the degree of inflationary pressures in the economy. A higher than expected reading should be taken as positive/bullish for the USD, while a lower than expected reading should be taken as negative/bearish for the USD. -Historical Impact on Assets: EURUSD USDJPY
Forecast:				
Previous:	50.30			
Time (GMT):	15:00			
Release Date	Actual	Forecast	Previous	
02-Dec-24	50.3	55.2	54.8	
01-Nov-24	54.8	49.9	48.3	
01-Oct-24	48.3	53.5	54.0	
03-Sep-24	54.0	52.1	52.9	
01-Aug-24	52.9	51.9	52.1	
01-Jul-24	52.1	55.8	57.0	

USD ISM Manufacturing Prices (Dec)
EURUSD

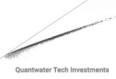


C:EURUSD Price Action - 2024-07-01 14:00 GMT



USD ISM Manufacturing Prices (Dec) | Actual: 52.1 | Forecast: 55.8 | Previous: 57.0

C:EURUSD Price Action - 2024-08-01 14:00 GMT

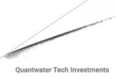


USD ISM Manufacturing Prices (Dec) | Actual: 52.9 | Forecast: 51.9 | Previous: 52.1

C:EURUSD Price Action - 2024-09-03 14:00 GMT



C:EURUSD Price Action - 2024-10-01 14:00 GMT



USD ISM Manufacturing Prices (Dec) | Actual: 48.3 | Forecast: 53.5 | Previous: 54.0

C:EURUSD Price Action - 2024-11-01 14:00 GMT



C:EURUSD Price Action - 2024-12-02 15:00 GMT



USD ISM Manufacturing Prices (Dec)
USDJPY



C:USDJPY Price Action - 2024-07-01 14:00 GMT



Price

- ✖ Event Time
- - - +1 Volatility Unit
- - - -1 Volatility Unit
- ✕ Max Peak
- ✕ Min Peak

ATR: 0.0564
Spread: 0.09900
Pre-ATR Volatility: 0.1200

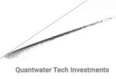
Volatility Unit (ATR*1.5+spread): 0.1835
Max Movement: 2.26x
Min Movement: 2.06x

Volume

12:00 12:30 13:00 13:30 14:00 14:30 15:00 15:30 16:00

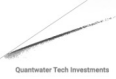
USD ISM Manufacturing Prices (Dec) | Actual: 52.1 | Forecast: 55.8 | Previous: 57.0

C:USDJPY Price Action - 2024-08-01 14:00 GMT



USD ISM Manufacturing Prices (Dec) | Actual: 52.9 | Forecast: 51.9 | Previous: 52.1

C:USDJPY Price Action - 2024-09-03 14:00 GMT



Price

- ✖ Event Time
- - - +1 Volatility Unit
- - - -1 Volatility Unit
- ✕ Max Peak
- ✕ Min Peak

ATR: 0.1034
Spread: 0.09900
Pre-ATR Volatility: 0.3160

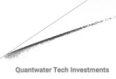
Volatility Unit (ATR*1.5+spread): 0.2541
Max Movement: 3.16x
Min Movement: 0.00x

Volume

12:00 12:30 13:00 13:30 14:00 14:30 15:00 15:30 16:00

USD ISM Manufacturing Prices (Dec) | Actual: 54.0 | Forecast: 52.1 | Previous: 52.9

C:USDJPY Price Action - 2024-10-01 14:00 GMT



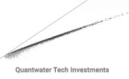
USD ISM Manufacturing Prices (Dec) | Actual: 48.3 | Forecast: 53.5 | Previous: 54.0

C:USDJPY Price Action - 2024-11-01 14:00 GMT



USD ISM Manufacturing Prices (Dec) | Actual: 54.8 | Forecast: 49.9 | Previous: 48.3

C:USDJPY Price Action - 2024-12-02 15:00 GMT



USD ISM Manufacturing Prices (Dec) | Actual: 50.3 | Forecast: 55.2 | Previous: 54.8

Crude Oil Inventories



Crude Oil Inventories

Event	Crude Oil Inventories			Description
Actual:				The Energy Information Administration's (EIA) Crude Oil Inventories measures the weekly change in the number of barrels of commercial crude oil held by US firms. The level of inventories influences the price of petroleum products, which can have an impact on inflation. If the increase in crude inventories is more than expected, it implies weaker demand and is bearish for crude prices. The same can be said if a decline in inventories is less than expected. If the increase in crude is less than expected, it implies greater demand and is bullish for crude prices. The same can be said if a decline in inventories is more than expected. -Historical Impact on Assets: USOILRoll
Forecast:	-			
Previous:	-4.237M			
Time (GMT):	16:00			
Release Date	Actual	Forecast	Previous	
27-Dec-24	-4.237	-0.700	-0.934	
18-Dec-24	-0.934	-1.600	-1.425	
11-Dec-24	-1.425	-1.000	-5.073	
04-Dec-24	-5.073	-1.600	-1.844	
27-Nov-24	-1.844	-1.300	0.545	
20-Nov-24	0.545	0.400	2.089	

Crude Oil Inventories
USOILRoll



USOILRoll Price Action - 2024-11-20 17:30 UTC+2 (CEST)

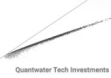


USOILRoll Price Action - 2024-11-27 17:30 UTC+2 (CEST)



Crude Oil Inventories | Actual: -1.844 | Forecast: -1.300 | Previous: 0.545

USOILRoll Price Action - 2024-12-04 17:30 UTC+2 (CEST)



USOILRoll Price Action - 2024-12-11 17:30 UTC+2 (CEST)



Crude Oil Inventories | Actual: -1.425 | Forecast: -1.000 | Previous: -5.073

USOILRoll Price Action - 2024-12-18 17:30 UTC+2 (CEST)



USOILRoll Price Action - 2024-12-27 20:00 UTC+2 (CEST)



Appendix: Key Economic & Analytical Concepts

Economic Data Interpretation

- **Forecast Vs. Actual:** Comparing predicted values with released data
- **Previous Reading:** Last reported value, used as a benchmark
- **Market Impact:** How releases affect financial markets

Average True Range (ATR)

- **Definition:** A technical indicator that measures market volatility
- **Calculations:**
 - Calculate True Range (TR) for each period:
 - $TR = \max [(high - low), abs(high - previous\ close), abs(low - previous\ close)]$
 - $ATR = \text{Simple moving average of TR over } N \text{ periods (typically 14)}$
- **Interpretation:**
 - Higher ATR = Higher Volatility
 - Lower ATR = Lower Volatility
- **Usage in Our Analysis:**
 - Helps standardize price movements across different market conditions
 - Forms the basis of our Volatility Unit Calculations

Volatility Unit

- **Formula:** $(ATR * multiple) + Spread$ (multiple is typically 1.5)
- **Purpose:** Standardizes price movements

Price Movements

- **Max Movement:** Largest upward price move after event
- **Min Movement:** Largest downward price move after event
- Both measured in Volatility Units

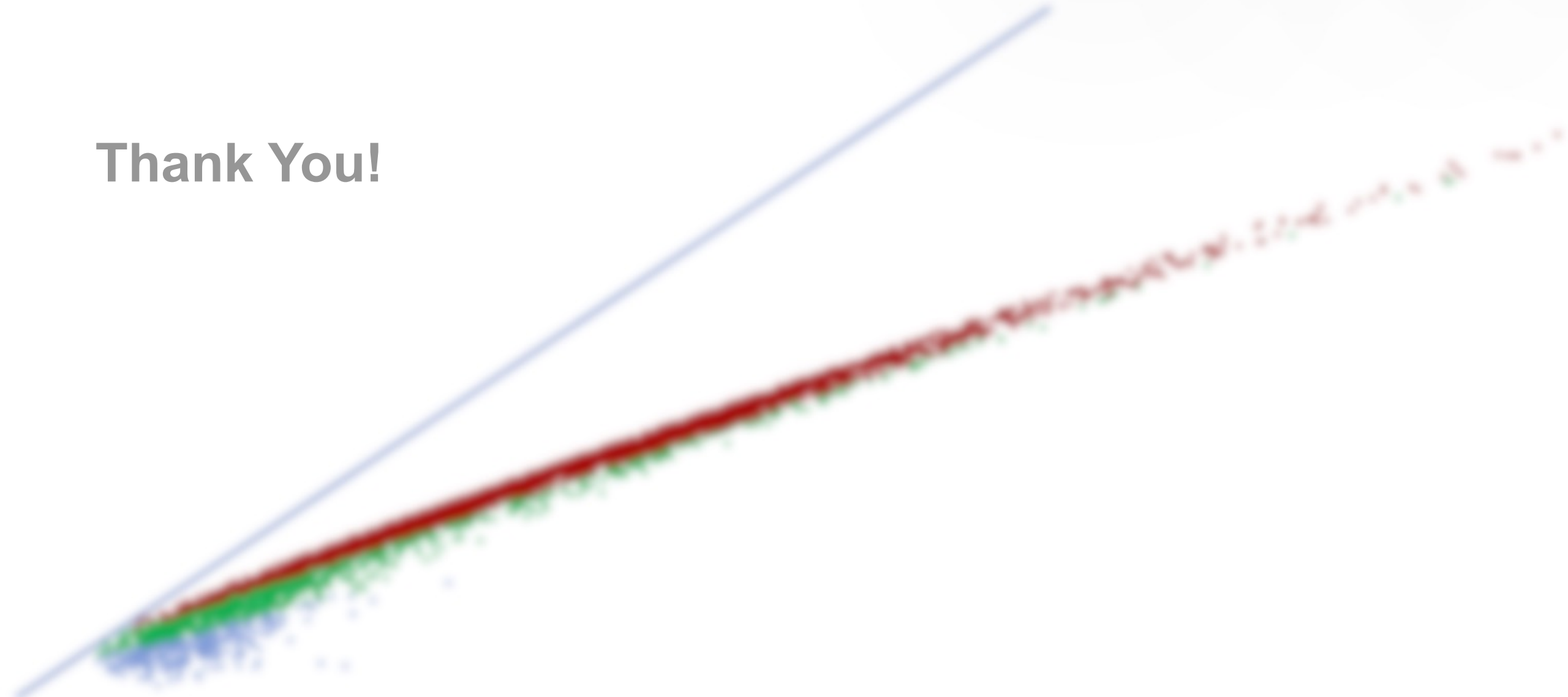
Using This Analysis

- Analyze volatility across different economic releases
- Identify patterns in market reactions
- Assess importance of economic indicators based on market impact

Resources

- <https://github.com/pawan-pro/quantwater-tech-investments/blob/main/RiskEventTrader>
- <https://quantwater.tech/research>

Thank You!



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